

Retinal support and barriers

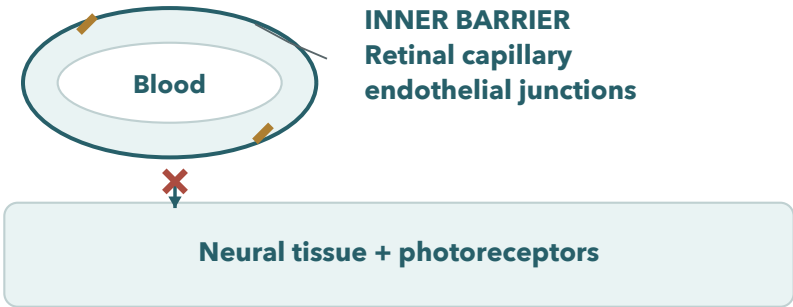
Protect neural tissue, maintain its chemical environment and renew the photoreceptor interface.

Two blood-retinal barriers

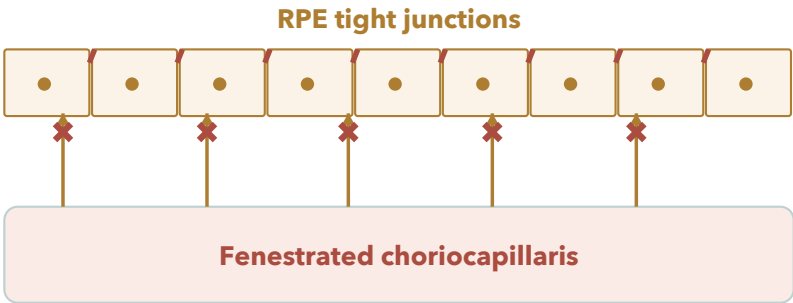
Which cell makes the seal?

VITREOUS SIDE

INNER NEURAL RETINA



OUTER BARRIER



Crosses show uncontrolled leakage blocked; selective cellular transport continues. Choroidal capillaries are fenestrated, so RPE supplies the outer seal.

Tracer experiment

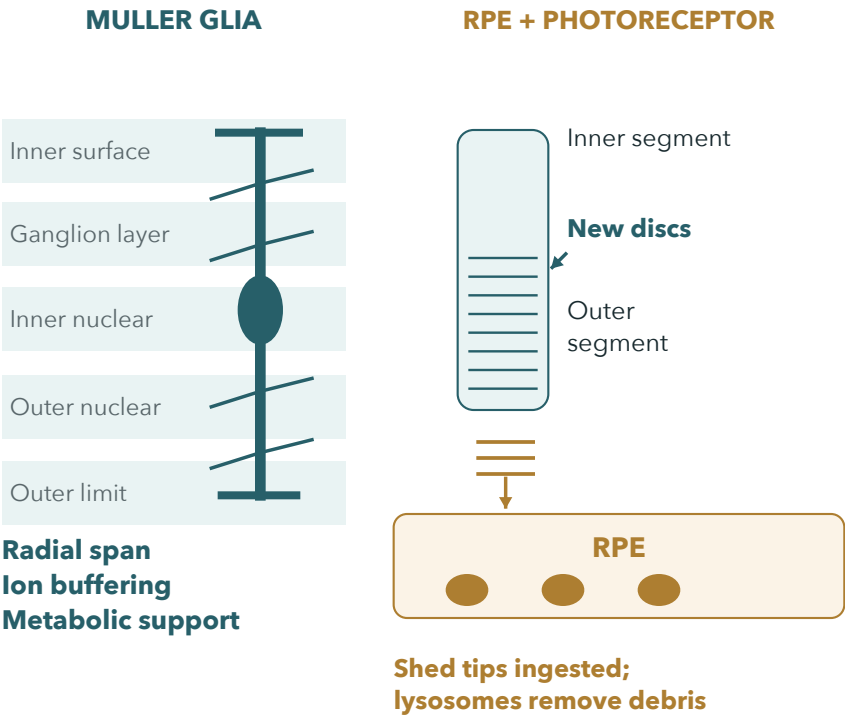
Tracer origin	Failed barrier site
Retinal capillary to neural retina	Inner barrier: retinal vascular endothelial junctions.
Choroid to subretinal compartment	Outer barrier: RPE junctions.

Keep the compartments distinct

The blood-retinal barriers protect retinal tissue. Nonpigmented ciliary epithelium and iris vascular endothelium belong to the blood-aqueous barrier.

Support cells, different jobs

A radial support cell and a renewal partner



Supporting cells preserve the environment for neurons.

Muller glia span neural-retinal layers. RPE lies outside neural retina and ingests shed photoreceptor outer-segment material.

Predict the consequence

Process lost	Likely problem
Muller ionic/metabolic support	Disturbed extracellular environment across retinal neurons.
RPE disc phagocytosis	Accumulating outer-segment debris and impaired renewal.
Pigment absorption	More internal stray light and reduced image contrast.

Why the central fovea is avascular

Removing vessels and displacing overlying neural layers reduces optical obstruction and scatter over densely packed cones. Avascularity does not mean the tissue receives no nutrition.